

Experiment7

- 1) Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int global_var = 10; // Global variable

void code_function()
{
    printf("Code address : %p\n", (void *)code_function);
}

int main() {
    static int static_var = 20; //
    Static variable    int stack_var = 30;    //
    Stack variable
    int *heap_var = malloc(sizeof(int)); // Heap variable

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
    }
    return 1;
}

*heap_var = 40;

printf("Linux Process Address Space\n");    printf("----
-----\n");

code_function();

printf("Global address : %p\n", (void *)&global_var);
printf("Static address : %p\n", (void *)&static_var);
printf("Heap address : %p\n", (void *)heap_var);
printf("Stack address : %p\n", (void *)&stack_var);
```

```
    free(heap_var);

    return 0;
}
```

OUTPUT :

```
charith@DESKTOP-RBQQSS:~$ nano exp7.1.c
charith@DESKTOP-RBQQSS:~$ gcc exp7.1.c -o exp7.1
charith@DESKTOP-RBQQSS:~$ ./exp7.1
Linux Process Address Space
-----
Code address : 0x5fb8dcc8a1c9
Global address : 0x5fb8dcc8d010
Static address : 0x5fb8dcc8d014
Heap address : 0x5fb90cb0a010
Stack address : 0x7ffd79371d6c
charith@DESKTOP-RBQQSS:~$ |
```

2) Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

int global_var = 100;

int main()
{
    static int static_var = 200;

    int stack_var = 300;
```

```
int *heap_var = malloc(sizeof(int));

*heap_var = 400;

printf("Process ID (PID): %d\n", getpid());

printf("Global variable address : %p\n", (void *)&global_var);

printf("Static variable address : %p\n", (void *)&static_var);

printf("Heap variable address  : %p\n", (void *)heap_var);

printf("Stack variable address : %p\n", (void *)&stack_var);

printf("\nProcess is running...\n");   getchar();

free(heap_var); return 0;

}
```

OUTPUT:

```
charith@DESKTOP-RBQQSS:~$ nano exp7.2.c
charith@DESKTOP-RBQQSS:~$ gcc exp7.2.c -o exp7.2
charith@DESKTOP-RBQQSS:~$ ./exp7.2
Process ID (PID): 1259
Global variable address : 0x5a9725068010
Static variable address : 0x5a9725068014
Heap variable address : 0x5a972bbfc010
Stack variable address : 0x7ffff5b390bc

Process is running...
```